

INTERNSHIP APPROVAL FORM

Student's full name: Lê Hải Trung

Major: Mechanical Engineering

Student ID: V202301000

Phone: 0332915559

Email: 23trung.lh@vinuni.edu.vn

Host company for internship: Phenikaa-X Join Stock Company, Phenikaa Group

Main business areas: Technology, Electronics, Autonomous Vehicle

Number of employees: 200

Company website: <https://phenikaa-x.com>

Industry supervisor's full name: Tran Trung Duc

Title: Head of UAV Department

Department: Phenikaa-X University, Yen Nghia, Ha Dong, Hanoi

Email: ductt@phenikaa-x.com

Phone: +84387217633

Details of the internship schedule: From 01/07/2025 to 31/10/2025

Key Date	Key Learning Activities
Day 1	Introduction to the internship program, rules, and equipment orientation
Week 1 & 2	Learn about quadrotors, hardware & software systems at PNKX; basic control using PX4
Week 3 & 4	Get familiar with Linux, install Ubuntu 22 & ROS2, practice with Turtlesim
Week 5	Install PX4, simulate flying with Gazebo, study MAVLink & mavros integration
Week 6&7	OFFBOARD mode control (outdoor): takeoff, circular & figure-8 flight (sim & real test)
Week 8	Study PX4's no-GPS mode and alternative positioning methods
Week 9	Install & use Indoor Positioning System (IPS) or AprilTag for indoor localization
Week 10&11	OFFBOARD mode control (indoor): hold position, circular & figure-8 flight (sim & real)
Week 12	Precision Landing control indoor using AprilTags
Week 13	Obstacle avoidance and collision prevention using 2D Lidar (Sim)
Week 14&15	Path planning problem using OFFBOARD control with Raspberry Pi
Week 16	Finalize report, prepare for end-of-term seminar
Final Day	Present outcomes, submit Git repo, and user manuals

Other important notes on the offered internship program:

Student's commitment

Confirmation of the host company (if necessary)

Approval of CECS

Date: 23/06/2025

Date 23/06/2025

Date

Signature: Lê Hải Trung

Signature: Trần Trung Đức

Signature

